

Gaussian small-contrast uniqueness and convexity for the maximal-energy density problem

Candidate partial-result packet for arXiv:1806.07225

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Status. Candidate substantial partial result, likely valid, awaiting specialist review. Both conclusions of Conjecture 3.8 hold for the Gaussian kernel on a bounded convex body at every fixed high-density volume fraction when the density contrast is sufficiently small.

1 Source conjecture

Osting and Simanek consider

$$E[\rho] = \frac{1}{2} \int_{\Omega \times \Omega} k(x, y) \rho(x) \rho(y) dx dy$$

over densities of mass one satisfying $\rho_- \leq \rho \leq \rho_+$. Their Conjecture 3.8 asks whether convexity of Ω and positivity of ρ_- force uniqueness of the maximizer and convexity of its high-density region B_+ [1].

$$= \int_{\Omega_0} \int_{\Omega} k(|x-y|) \rho_-(y) dy + \int_{\Omega_\varepsilon \setminus \Omega_0} \int_{\Omega} k(|x-y|) \rho_-(y) dy - \int_{\Omega_0 \setminus \Omega_\varepsilon} \int_{\Omega} k(|x-y|) \rho_-(y) dy$$

be independent of ε . But, this is false if f is convex, decreasing, and positive. To see this, we observe that as ε changes the change in the values of $K\rho^*(x)$ at $x = (r', 0)$ and $x = (0, r')$ have opposite sign. For $x = (r', 0)$, the value of $K\rho^*(x)$ is decreasing in ε since

$$\frac{f(1-r') + f(1+r')}{2} \geq f(1) \geq f\left(\sqrt{1+(r')^2}\right).$$

The first inequality follows from the convexity of f and the second from f being decreasing. Similarly, for $x = (0, r')$, the value of $K\rho^*(x)$ is increasing in ε . This shows that for all domains Ω_ε with $\varepsilon > 0$ sufficiently small, the region where $\rho^* \equiv \rho_+$ is not a ball. However, we believe that as $\rho_+ \rightarrow \infty$, the region where $\rho^* \equiv \rho_+$ converges to a shrinking ball.

In light of the above examples and observations, we make the following conjecture.

Conjecture 3.8. *Suppose Ω and k satisfy the assumptions in Section 1. Suppose also that Ω is convex and $\rho_- > 0$. Then the optimal density for (4) is unique and B_+ is convex.*

Our next example shows that in general, convexity is not needed to deduce uniqueness of the optimal density.

3.6. Example: the cross. Let us consider the case when $\Omega \subset \mathbb{R}^2$ and is given by

Figure 1: Conjecture 3.8 on PDF page 10 of the source paper.

2 Proof intuition

At small contrast the energy is led by a linear “bathtub” problem. Its potential is the convolution of the convex body’s indicator with a Gaussian, whose level sets have strictly positive curvature. The nonlinear correction does not merely give convergence to this reference set: the KKT condition defines a threshold map on fixed-volume sets, and that map is a contraction. The unique fixed point is therefore the unique maximizer. Its threshold surface is a small C^2 perturbation of a strictly convex surface, hence stays convex. The argument does not control arbitrary contrast.

3 The partial theorem

Let $\Omega \subset \mathbb{R}^d$ be a bounded convex body, let $M = |\Omega|$ and $c = M^{-1}$, and fix $\tau > 0$ and $\theta \in (0, 1)$. Put

$$G_\tau(z) = (4\pi\tau)^{-d/2} e^{-|z|^2/(4\tau)}, \quad k_\tau(x, y) = G_\tau(x - y).$$

For $\varepsilon > 0$ define

$$\rho_-^\varepsilon = c - \theta\varepsilon, \quad \rho_+^\varepsilon = c + (1 - \theta)\varepsilon.$$

Theorem 1. *For every bounded convex body Ω , $\tau > 0$, and $\theta \in (0, 1)$, there is $\varepsilon_0 > 0$ such that, whenever $0 < \varepsilon < \varepsilon_0$, the Gaussian energy problem with bounds $\rho_-^\varepsilon, \rho_+^\varepsilon$ has a unique maximizer modulo null sets. It has the form*

$$\rho_\varepsilon = c + \varepsilon(\mathbf{1}_{B_\varepsilon} - \theta), \quad |B_\varepsilon| = \theta M,$$

and B_ε is convex modulo null sets.

The Gaussian is explicitly listed among the source’s model kernels. There is a wording caveat: $e^{-r^2/(4\tau)}$ is completely monotone in r^2 , not in r . Thus Theorem 1 addresses the source’s explicitly intended Gaussian subclass even if its preceding condition is read literally.

4 Proof

Define the container potential

$$V(x) = (G_\tau * \mathbf{1}_\Omega)(x) = \int_\Omega G_\tau(x - y) dy.$$

Step 1: strict curvature of Gaussian-potential levels

Let μ_x be the probability measure on Ω with density $G_\tau(x - y)/V(x)$. Differentiation under the integral gives

$$\nabla^2 \log V(x) = -\frac{1}{2\tau} I + \frac{1}{4\tau^2} \text{Cov}_{\mu_x}(Y). \quad (1)$$

The Brascamp–Lieb covariance inequality gives $\text{Cov}_{\mu_x}(Y) \preceq 2\tau I$ [2]. Equality in a nonzero direction would require an untruncated Gaussian direction; this is impossible because Ω is bounded. Hence the inequality in (1) is negative definite. On every compact level surface of V , compactness gives a uniform negative bound.

In particular, V has a unique maximum, every other level is noncritical, and every positive superlevel of V has strictly convex smooth boundary. For the unique number t_θ satisfying

$$|\Omega \cap \{V > t_\theta\}| = \theta M,$$

the reference bathtub set $B_0 = \Omega \cap \{V > t_\theta\}$ is convex.

Step 2: the threshold map

Let $\mathcal{B}_\theta$ be the measurable subsets of Ω of measure θM , modulo null sets, with metric $d(B, C) = |B \triangle C|$. This metric space is complete. For $B \in \mathcal{B}_\theta$ set

$$\rho_B = c + \varepsilon(\mathbf{1}_B - \theta), \quad P_B = G_\tau * \rho_B = cV + \varepsilon G_\tau * (\mathbf{1}_B - \theta \mathbf{1}_\Omega).$$

Let λ_B be the fixed-volume quantile of P_B and define

$$T(B) = \Omega \cap \{P_B > \lambda_B\}, \quad |T(B)| = \theta M.$$

Gaussian convolution is analytic and nonconstant, so its level sets have zero Lebesgue measure and the definition is unambiguous.

For $B, C \in \mathcal{B}_\theta$,

$$\|P_B - P_C\|_\infty \leq \varepsilon \|G_\tau\|_\infty |B \triangle C|. \quad (2)$$

Fixed-volume quantiles are 1-Lipschitz under uniform perturbations, so $|\lambda_B - \lambda_C|$ is bounded by the right side of (2). Uniformly in B , P_B is C^2 -close to cV as $\varepsilon \downarrow 0$. The selected levels therefore stay in a compact noncritical band around ct_θ . The coarea formula supplies constants $A, s_0 > 0$, independent of B , such that

$$|\{x \in \Omega : |P_B(x) - \lambda_B| \leq s\}| \leq As \quad (0 < s < s_0). \quad (3)$$

If $x \in T(B) \triangle T(C)$, the two threshold inequalities and (2) place x in one of the bands in (3) with width at most twice the right side of (2). Consequently

$$|T(B) \triangle T(C)| \leq 2A\varepsilon \|G_\tau\|_\infty |B \triangle C|.$$

For sufficiently small ε , T is a strict contraction and has a unique fixed point B_ε .

Step 3: the fixed point is the unique maximizer

The Gaussian kernel is strictly positive definite, so the quadratic energy is strictly convex as a functional of ρ . A maximizer over the bounded mass class is therefore bang–bang, hence is ρ_B for some $B \in \mathcal{B}_\theta$. The usual mass-transfer first variation says that high density can occur only on the top fixed-measure values of P_B ; otherwise moving equal mass from a lower-potential high-density piece to a higher-potential low-density piece raises the energy. Thus every maximizer satisfies $B = T(B)$ modulo null sets. The contraction has only one fixed point, proving uniqueness.

Finally, P_B is uniformly C^2 -close to cV . The full selected superlevel surface $\{P_B = \lambda_B\}$ is a small normal perturbation of the strictly convex surface $\{V = t_\theta\}$. Its principal curvatures remain positive. Hence $\{P_B > \lambda_B\}$ is convex, and so is its intersection with the convex body Ω . Applying this to the fixed point proves that B_ε is convex and completes the proof. $\square$

5 Scope, checks, and novelty

The theorem proves both conclusions of Conjecture 3.8 for every fixed volume fraction in a sufficiently small Gaussian density-contrast regime. It does not settle arbitrary contrast or the source’s entire kernel class. The same contraction proof applies to any smooth kernel for which the selected container-potential level has uniform positive curvature and convolution gives a uniform C^2 perturbation bound.

A bounded search through 13 August 2026 used the exact title and conjecture, together with combinations of *Gaussian kernel*, *bang–bang density*, *small contrast*, *unique optimizer*, *convex superlevel*, and *threshold map*. It found the standard log-concavity and nonlocal-perimeter literature but no exact result for this problem. Novelty confidence is moderate because the ingredients are classical. The main reviewer targets are the uniform coarea estimate at a nonsmooth container boundary and strictness in the truncated-Gaussian covariance inequality.

References

- [1] B. Osting and B. Simanek, *A maximal energy pointset configuration problem*, arXiv:1806.07225 (2018); J. Math. Anal. Appl. 485 (2020), 123830.
- [2] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation*, J. Funct. Anal. 22 (1976), 366–389.